Slopes and local change

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Slopes and local change before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Loss functions

Without opening the old lesson, explain one key idea from Loss functions. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Forward passes by hand

Without opening the old lesson, explain one key idea from Forward passes by hand. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Bias terms and decision boundaries

Without opening the old lesson, explain one key idea from Bias terms and decision boundaries. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Lesson reference for workbook practice

> Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

Let a one-example model predict w and let its target be 2. Its squared loss is L(w) = (w - 2) squared. Here w is the parameter we vary, not a new data example. L is a nonnegative function of that parameter.

w	Prediction minus target	L(w)
0	-2	4
1	-1	1
2	0	0
3	1	1

w	Prediction minus target	L(w)
4	2	4

From w = 0 to w = 1, the loss changes by -3 while w changes by 1. The average **slope** is -3. From w = 1 to w = 2 it is -1. The curve does not have one fixed **slope** everywhere.

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Slopes and local change trace and record the first step you can explain without help.

Use W2: Check a symmetric estimate for **Slope**s and **local change**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Slopes and local change explanation against the supplied facts before discussing it.

Use the worked case for **Slope**s and **local change**. Proposed explanation: “Divide by the step only”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Estimate a new curve independently

Let J(a) = (a + 1) squared. At a = 1 use step 0.25 to find the symmetric **slope**. Compare J(1) with J(0.9), then explain why moving to -4 does not follow safely from the **slope** clue.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Measure an interval slope

> Measure an interval slope. Apply the supplied Slopes and local change facts and keep your reasoning.

For L(w) = (w - 2) squared, calculate the average **slope** from w = 0 to w = 2. Compare it with the local **slope** at w = 2.

My reasoning and evidence

WORKBOOK · 6 OF 15

W2 Check a symmetric estimate

> Check a symmetric estimate. Apply the supplied Slopes and local change facts and keep your reasoning.

At centre w = 0, use step 0.5. Calculate L(-0.5), L(0.5) and the symmetric **slope**.

My reasoning and evidence

WORKBOOK · 7 OF 15

W3 Challenge a large step

> Challenge a large step. Apply the supplied Slopes and local change facts and keep your reasoning.

Starting at w = 3, compare loss after moving to 2.9 and after moving to -1. Both moves decrease w.

My reasoning and evidence

WORKBOOK · 8 OF 15

W4 Estimate a new curve independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Estimate a new curve independently. Apply the supplied Slopes and local change facts and keep your reasoning.

Let J(a) = (a + 1) squared. At a = 1 use step 0.25 to find the symmetric **slope**. Compare J(1) with J(0.9), then explain why moving to -4 does not follow safely from the **slope** clue.

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Limit zero slope evidence

Limit zero slope evidence. Apply the supplied Slopes and local change facts and keep your reasoning.

For F(w) = (w squared - 1) squared, compare F(0) and F(0.1). Does zero local **slope** at zero prove a minimum?

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

Connect Slopes and local change to Image-pattern Network Investigation, identify a limit and choose the next useful test.

Add a labelled **slope**s and **local change evidence** sheet to Image-pattern Network Investigation. Show how you can calculate an average rate of change over a stated interval.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: Slopes and local change

> Use a local slope to propose a small move rather than claiming every move helps. Explain the labelled step before checking the changed condition.

Use a local **slope** to propose a small move rather than claiming every move helps.

Toy loss $J(a)=(a-2)^2$. At $a = 0$, use symmetric step 0.5.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/9

Slopes and local change

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Evaluate nearby points

J(0.5)=2.25 and J(-0.5)=6.25.



2. Estimate slope

Slope=(2.25-6.25)/(0.5-(-0.5))=-4.



3. Check small move

Move to a = 0.25: loss 3.0625, below starting loss 4.

CHANGE ONE THING

Move from 0 all the way to5.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Which calculation disproves “any increase helps”? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Image-pattern Network Investigation record.

Project: Image-pattern Network Investigation

Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Use a local **slope** to propose a small move rather than claiming every move helps.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Move to a = 0.25: loss 3.0625, below starting loss 4.

1. Check small move

2. Estimate **slope**

3. Evaluate nearby points

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Check small move Move to a = 0.25: loss 3.0625, below starting loss 4. A **slope** describes how the loss changes near the current parameter.

The symmetric calculation compares equal small moves on either side. Its negative result suggests a small increase may help at zero.

It does not promise every increase helps: jumping to five passes the optimum at two and produces a larger loss. Check the new loss after each proposed step.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Move from 0 all the way to5. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. Loss becomes 9, exceeding the starting 4.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Loss becomes 9, exceeding the starting 4. A negative local **slope** suggests increasing a locally, not that every positive jump decreases loss.

Repair a missing building block

Review Loss functions (ai-5/8). Goal: Calculate per-example squared error and mean squared error. Short practice: Why must the report name the chosen loss?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Forward passes by hand (ai-5/6). Goal: Record scores and activations in dependency order. Short practice: Why should feedback identify h before correcting only y?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Bias terms and decision boundaries (ai-5/3). Goal: Calculate a weighted score with a bias term. Short practice: Is B positive under the original or changed model?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

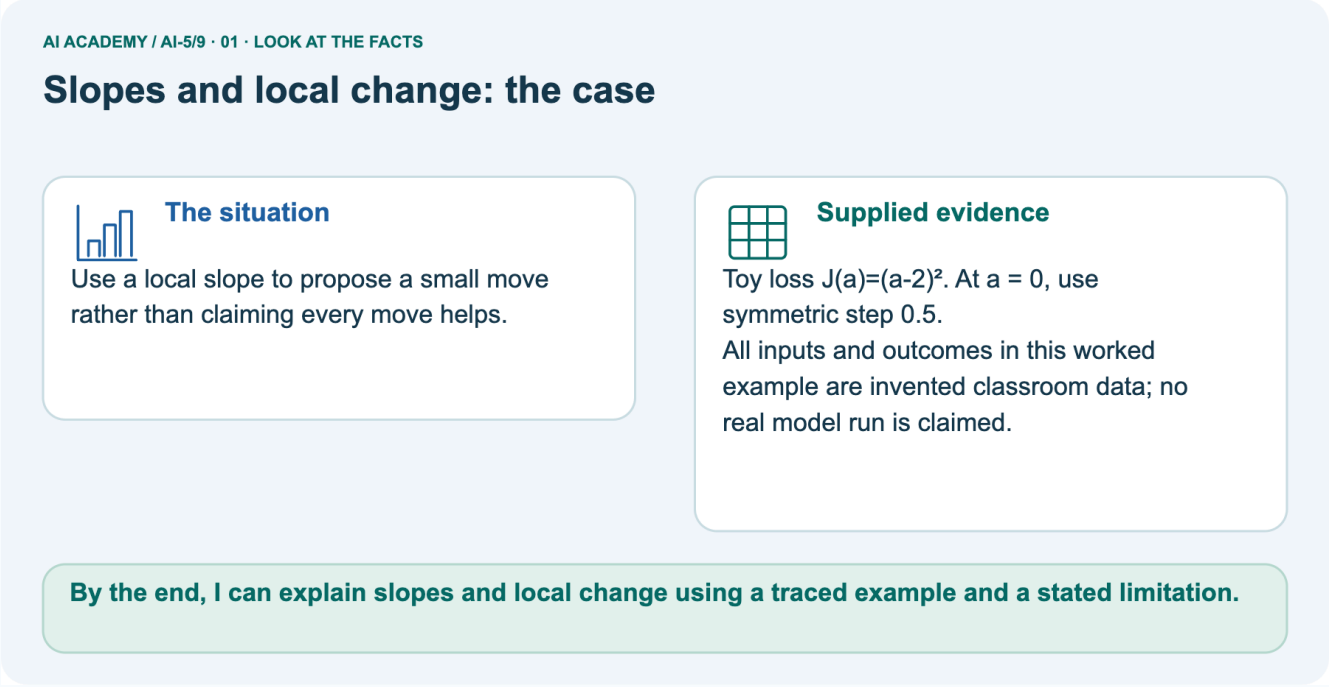
Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**

Visual concept journey

> Explain each Slopes and local change picture from its labels, then record a relevant limit.

01. Slopes and local change: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

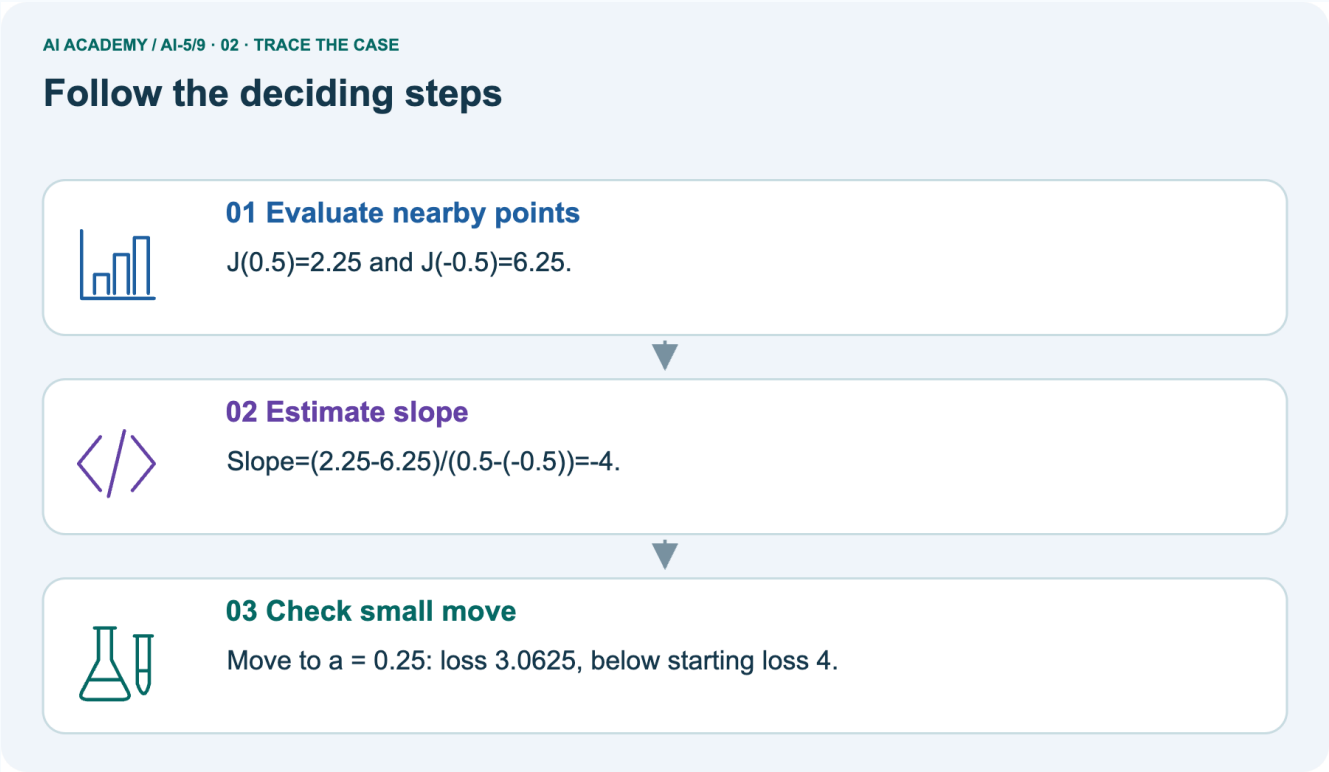
Use a local **slope** to propose a small move rather than claiming every move helps. Supplied facts: Toy loss J(a)=(a-2)². At a = 0, use symmetric step 0.5.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Evaluate nearby points: J(0.5)=2.25 and J(-0.5)=6.25. → Estimate **slope**: **Slope**=(2.25-6.25)/(0.5-(-0.5))=-4. → Check small move: Move to a = 0.25: loss 3.0625, below starting loss 4.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Compare slope clues on the same loss curve

AI ACADEMY / AI-5/9 · 03 · INSPECT THE LESSON MODEL

Compare slope clues on the same loss curve

Current w	Local slope	Nearby downhill direction
1	-2	Increase w a little
2	0	No first order direction
3	2	Decrease w a little

These are local clues for L(w) = (w - 2) squared. At w = 2 the displayed function has its minimum; zero slope does not establish a minimum for every possible function.

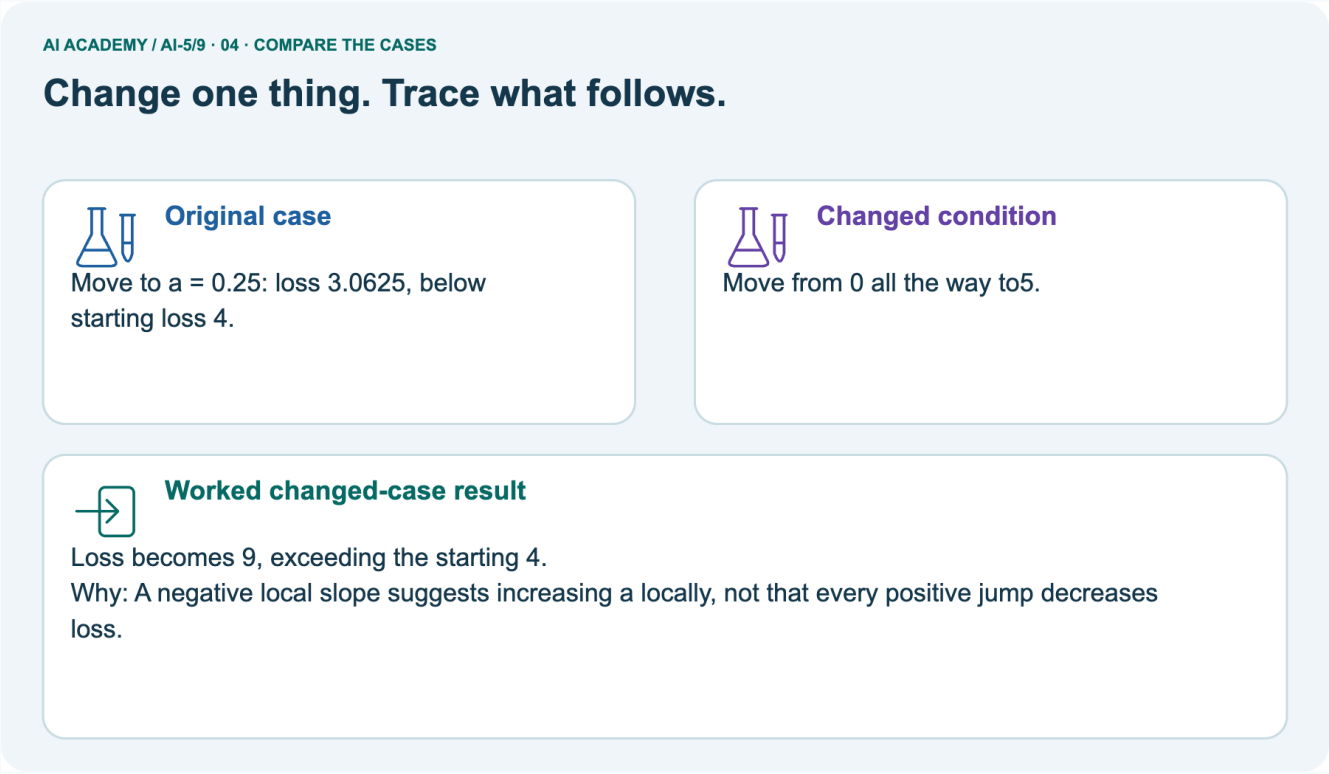
Follow the labelled picture with the matching explanation. Open the vector diagram

Compare **slope** clues on the same loss curve. 1 | -2 | Increase w a little. 2 | 0 | No first order direction. 3 | 2 | Decrease w a little

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

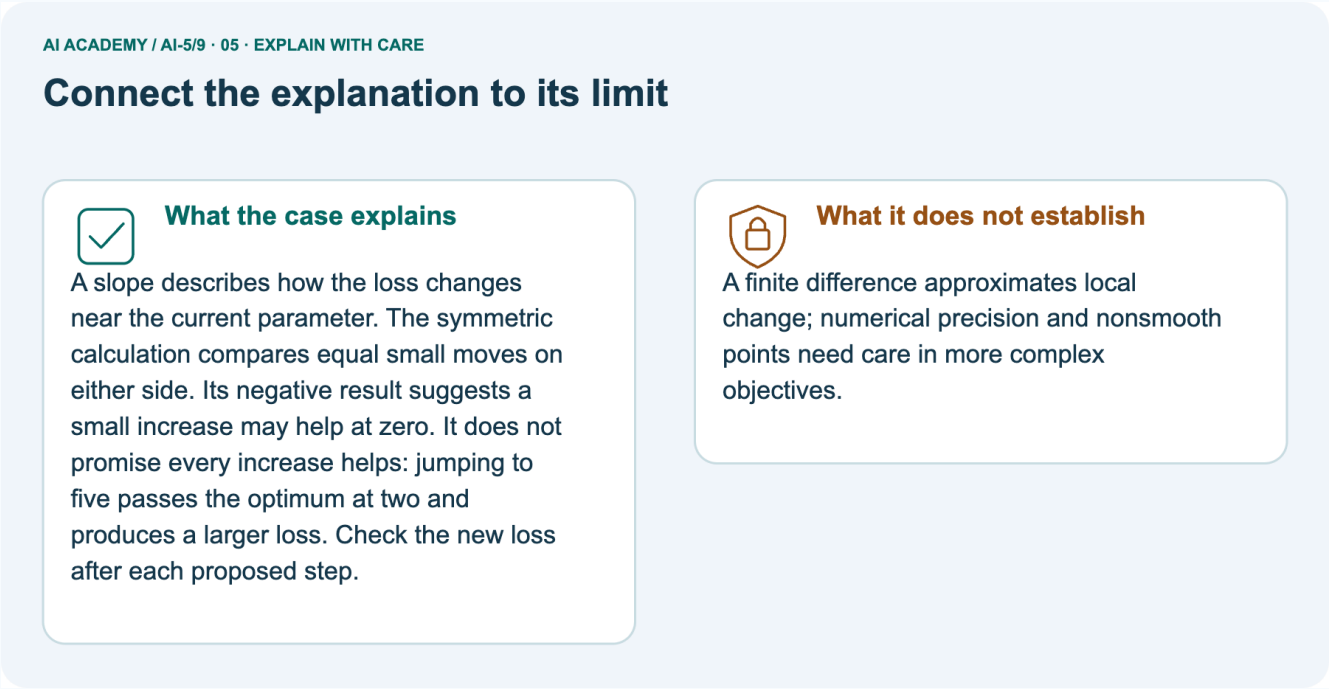
Original case: Move to a = 0.25: loss 3.0625, below starting loss 4. Changed condition: Move from 0 all the way to5. Expected result: Loss becomes 9, exceeding the starting 4.

Reason: A negative local **slope** suggests increasing a locally, not that every positive jump decreases loss.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A **slope** describes how the loss changes near the current parameter. The symmetric calculation compares equal small moves on either side. Its negative result suggests a small increase may help at zero.

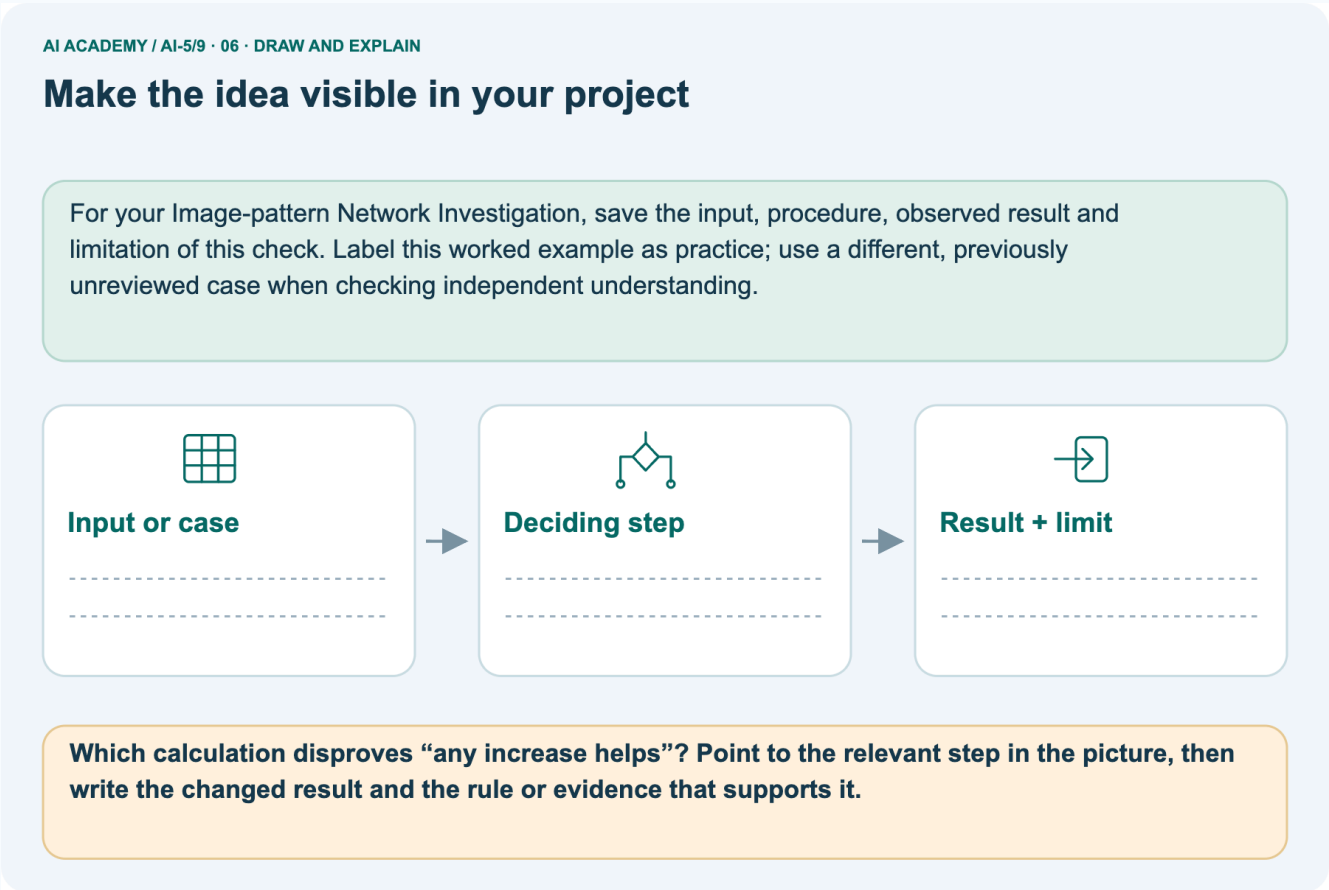
It does not promise every increase helps: jumping to five passes the optimum at two and produces a larger loss. Check the new loss after each proposed step.

Limit: A finite difference approximates **local change**; numerical precision and nonsmooth points need care in more complex objectives.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which calculation disproves “any increase helps”? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.